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Professional Profile

Experienced independent software engineer specialising in graphics, rendering, developer tooling, and Apple-platform development. Designed and implemented custom software systems from first principles throughout a 25+ year development career.

Extensive experience across the full software development lifecycle, from concept and UX design through to implementation, testing, App Store submission, and long-term support. Highly self-motivated.

Seeking a role as an iOS Engineer, macOS Developer, or Software Engineer specialising in Apple platforms.

Technical Skills

Programming Languages

- Swift, Objective-C, Metal Shading Language (MSL), C++

Frameworks & APIs

- SwiftUI, UIKit, AppKit, Core Animation, Core Graphics, Core Image
- Metal, MetalKit, AVFoundation, CloudKit, CoreText
- Grand Central Dispatch (GCD), Swift Concurrency, URLSession

Graphics & Rendering

- GPU rendering, compute shaders, real-time rendering
- Custom rendering engines, Metal compute/render pipelines
- Texture management

Tools & Technologies

- Xcode, Instruments, XCTest, Blender, Git, App Store Connect

Core Competencies

- iOS and macOS development (full application lifecycle)
- Cross-platform Apple development
- MVC and MVVM architecture
- Multithreading and concurrency
- Memory optimization, ARC/manual memory management
- Performance profiling and debugging
- UIKit and SwiftUI integration

- Asynchronous networking and background processing
- Custom UI controls and rendering systems
- App Store submission, release, and maintenance

Professional Experience

Self-Employed Software Developer | 1999 – Present

Designed and developed software, graphics technologies, developer tools, and Apple-platform applications over a 25+ year independent development career.

Expanded primary development stack to include Swift and SwiftUI, resulting in three App Store releases in 2026.

Selected Portfolio

app: **HowLS** (SwiftUI, 2026) c.f. <https://writtenbyabriton.com/HowLS/>

- Productivity application designed to help users manage recurring low-priority tasks without traditional deadline pressure.
- Developed and released from concept to App Store production in under two weeks, including design, implementation, testing, assets, and App Store materials.
- Source code released on GitHub.

app: **Block Ed** (Obj-C/MSL, 2025) c.f. <https://writtenbyabriton.com/pbgs-block-ed/>

- Designed and developed a universal Apple application supporting both macOS and iOS.
- Implemented CloudKit-based synchronization.
- Developed custom Metal shader effects and rendering pipelines.
- Built PDF export, undo/redo, autosave, and backup systems.
- Adapted user interfaces across multiple device classes and screen sizes.
- Localised application into four languages.

Notable Technical Contributions & Internal Tools (1999-2025)

project: **Video Generation Engine**

- Developed an automated content-generation system producing multilingual promotional and tutorial videos from plain-text scripts.
- Automated screenshot capture, asset integration, and video assembly workflows.

project: **Advanced Typography Library**

- Developed a custom text-rendering system supporting gradients, shadows, animated texture fills, and 3D extrusion effects.

project: **Handwriting Animation System**

- Animated handwriting generator using vector-based custom triangulation (rather than rasterisation) for smooth, variable stroke-width rendering.

project: **Logo Design & Export Tool**

- Template-based logo generator using Metal compute shaders to perform real-time rendering and rapid iteration.
- Automatically exports multiple asset variants and configuration files (.json) for single-step app integration.

Earlier Software Products and Technologies

- Developed a C++ WYSIWYG script editor featuring spell checking and booklet-printing support.
- Developed Windows and iOS games, including an arcade title and experimental audio-generation applications.
- Created an incremental-backup utility supporting compressed and versioned file storage.
- Built custom graphics, typography, cryptography, and document-generation libraries for reuse across multiple projects.

Education

- BA in Mathematics (1st Class Honours), University of York, 1997